TECHNICAL RESEARCH REPORT · SAKURA ANALYTICS

# Forecasting Japan's Cherry Blossom Full Bloom

Mathematical foundations, leakage-aware implementation and expanding-window validation of historical climatology and adapted accumulated-temperature baselines.

82 Somei-yoshino stations | Two forecast issue dates | Five CV folds | Validation + held-out test

01 / AT A GLANCE

## Executive summary

The 30-year station climatology is the strongest baseline for minimizing average date error. Regionally calibrated temperature rules improve substantially over a national threshold, but do not surpass climatology on MAE.

| 3.804 d | 48.48% | 11.0% | 575 |
|---|---|---|---|
| Best validation MAE<br>30-year climatology | Best ±3-day hit rate<br>Regional rule · Feb 1 | Feb 1 MAE reduction<br>Regional vs national rule | Validation station-years<br>per forecast date |

### Principal conclusions

**Historical benchmark:** preceding-30-year station mean, with a regional/national fallback for insufficient station history. **Temperature benchmark:** region-specific accumulated-maximum-temperature thresholds, estimated from training data. **Operational finding:** the March 1 temperature forecast did not systematically outperform February 1, despite observing February weather.

All metrics below are cross-validation results. Model selection and final independent testing on 2015–2025 have not been completed for the team's champion models.

02 / RESEARCH DESIGN

## Dataset, cutoffs & validation

The team provides standardized forecast-date CSVs in `data/model/`, loaded through `src/model_data.py`. Daily maximum temperatures come from the cleaned station-day weather series for the temperature-rule implementation.

**4,365**
Station-year rows in each issue-date CSV

**3,837**
Development rows through 2014

**528**
Held-out test rows, 2015–2025

**82**
Eligible Somei-yoshino stations

## Issue-date information boundary

**FEBRUARY 1**

### Earlier-season forecast

Current-season observations available through **January 31**. All February-and-later maximum temperatures must be estimated, not taken from realized future weather.

**MARCH 1**

### Updated forecast

Current-season observations available through **February 28 or 29**. Observed February maximum temperatures are combined with estimates for March onward.

## Expanding-window cross-validation

| Fold | Training years | Validation years | Validation N |
|---|---|---|---|
| 1 | 1964–2004 | 2005–2006 | 148 |
| 2 | 1964–2006 | 2007–2008 | 127 |
| 3 | 1964–2008 | 2009–2010 | 106 |
| 4 | 1964–2010 | 2011–2012 | 98 |
| 5 | 1964–2012 | 2013–2014 | 96 |

The five validation blocks contain 575 station-year observations in total. All models use the same fold boundaries; temperature threshold and weather climatology are recomputed within each training fold.

## Mathematics of evaluation

Let $y_i$ denote actual full-bloom day of year and $\hat{y}_i$ the point prediction, with error $e_i = \hat{y}_i - y_i$. Positive error means the prediction is late.

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i - y_i|$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

$$\mathrm{HitRate}_{\pm 3} = \frac{1}{n}\sum_i \mathbf{1}(|\hat{y}_i - y_i| \le 3)$$

Reported summary metrics are **unweighted means of the five fold-level metrics**, following the shared `cv_report()` convention. This differs from computing pooled MAE/RMSE/$R^2$ over all 575 observations; pooled and fold-mean results should not be interchanged.

**03 / BASELINE ONE**

# 30-year station climatology

The historical climatology predicts each station's full-bloom day of year from the preceding 30 *calendar* years, using only observations available in the current training fold.

$$\hat{y}_{s,t} = \frac{1}{N_{s,t}} \sum_{i \in H_{s,t}} y_{s,i}$$

$$H_{s,t} = \{ i \ : \ t - 30 \le i < t, \ i \in \text{Train}, \ s_i = s \}$$

Here **s** is the station, **t** the prediction year, and $\mathbf{N_{s,t}}$ the number of valid historical bloom observations in the window. Missing years are not replaced by observations older than the 30-year window.

### Fallback logic

1. **Station mean**
   Use the station's mean when at least 10 historical station observations are available.
2. **Regional mean**
   Otherwise use the training-only regional historical mean within the same 30-year window.
3. **National mean**
   Use the training-only national historical mean if the region has no eligible history.

The 10-observation minimum and fallback hierarchy are **baseline-specific implementation decisions**, not formal team-wide requirements. This model uses no current-season weather, so its February 1 and March 1 predictions are identical.

| Implementation | MAE (days) | RMSE | $R^2$ | ±3-day hit |
|---|---|---|---|---|
| All-history climatology | 3.956 | 4.687 | 0.804 | 44.00% |
| **30-year climatology** | **3.804** | **4.573** | **0.819** | **45.72%** |

The 30-year window reduces fold-mean MAE by approximately **3.8%** relative to the all-history version. Fold 5 (2013–2014) is the weakest, with MAE of **6.049 days**, demonstrating that historical averages cannot fully capture atypical seasons.

**04 / BASELINE TWO**

# Adapted 600°C temperature rule

The traditional 600°C heuristic accumulates **daily maximum temperature** from February 1 to estimate first flowering. This project targets **full bloom**, so the temperature threshold is recalibrated from historical full-bloom observations.

### Step A · Historical temperature accumulation

$$S_{s,y}(d) = \sum_{u=\text{Feb 1}}^{d} T^{\max}_{s,y,u}$$

For each eligible training station-year, sum the observed daily maximum temperatures through the *actual* full-bloom date. Only complete February 1-to-full-bloom sequences are eligible for calibration.

### Step B · Calibrate the full-bloom threshold

$$\theta_{\text{national},k} = \text{median}_{(s,y)\in \text{Train}_k}\left[S_{s,y}(d^{\text{full}}_{s,y})\right]$$

For the regional variant, calculate the same median separately for each region **r**:

$$\theta_{r,k} = \text{median}_{(s,y)\in \text{Train}_k,\ r(s)=r}\left[S_{s,y}(d^{\text{full}}_{s,y})\right]$$

The national threshold was **744.20°C** in Fold 1 (3,262 eligible calibration observations), increasing modestly to **748.20°C** by Fold 5. These are training-derived medians, **not** thresholds optimized directly for validation MAE.

| Fold | Training observations for calibration | National threshold (°C) |
|---|---|---|
| 1 | 3,262 | 744.20 |
| 2 | 3,410 | 744.55 |
| 3 | 3,537 | 746.30 |
| 4 | 3,643 | 747.40 |
| 5 | 3,741 | 748.20 |

### Step C · Estimate future daily temperature

For a validation year in fold **k**, estimate the maximum temperature for station **s** on calendar day **m** from the preceding 30 permitted training-weather years:

$$\hat{T}^{\max}_{s,m,k} = \frac{1}{|W_{s,m,k}|} \sum_{y \in W_{s,m,k}} T^{\max}_{s,y,m}$$

$$W_{s,m,k} \subseteq \{Y_k - 29, \ldots, Y_k\}$$

Here $\mathbf{Y_k}$ is the last year of fold **k**'s training window. The implementation uses a national day-of-year fallback when a station-specific estimate is unavailable; no national fallback days were recorded in the reported validation forecasts.

### Step D · Issue-date-safe forecast

$$\tilde{T}^{\max}_{s,y,u}(c) = T^{\max}_{s,y,u} \quad \text{if } u < c$$

$$\tilde{T}^{\max}_{s,y,u}(c) = \hat{T}^{\max}_{s,m(u),k} \quad \text{if } u \geq c$$

$$\hat{d}^{\text{full}}_{s,y}(c) = \min\left\{ d \geq c : \sum_{u=\text{Feb 1}}^{d} \tilde{T}^{\max}_{s,y,u}(c) \geq \theta_{r(s),k} \right\}$$

For **February 1**, the entire February–June trajectory is estimated. For **March 1**, February observed temperatures are included and the remainder is estimated. The actual algorithm accumulates from February 1 and finds the first crossing; the issue-date validity check confirms no reported regional predictions fell before issuance.

> **Interpretation:** A threshold of 744°C does *not* mean a single day reaches 744°C. It is the **sum** of daily maximum temperatures over many days. It is also not a Growing Degree Day (GDD) sum with a base-temperature subtraction.

## Regional calibration values

| Region | Historical median accumulation (°C) |
|---|---|
| Hokkaido | 569.6 |
| Tohoku | 662.5 |
| Chubu | 715.2 |
| Chugoku | 746.7 |
| Kansai | 761.9 |
| Kanto | 787.3 |
| Shikoku | 800.4 |
| Kyushu | 844.1 |

These eight values come from the regional training-calibration diagnostic shared during model development. They are not final test-set results, and regional thresholds are recalculated separately inside each CV fold.

**05 / VALIDATION EVIDENCE**

# Baseline performance comparison

All five configurations were evaluated under the team's shared fold boundaries and metric definitions. Hit rates below are displayed as percentages.

| Model | MAE ↓ | RMSE ↓ | $R^2$ ↑ | ±3-day hit ↑ |
|---|---|---|---|---|
| **30-year station climatology**<br>Historical | 3.804 | 4.573 | 0.819 | 45.72% |
| **National rule · Feb 1**<br>National | 4.818 | 6.304 | 0.632 | 45.33% |
| **Regional rule · Feb 1**<br>Regional | 4.287 | 5.406 | 0.743 | 48.48% |
| **National rule · Mar 1**<br>National | 4.975 | 6.527 | 0.607 | 44.71% |
| **Regional rule · Mar 1**<br>Regional | 4.383 | 5.532 | 0.726 | 48.23% |

## Mean absolute error by configuration




Mean of five fold-level MAEs. Lower is better.

HISTORICAL BENCHMARK

### Best MAE: 3.804 days

The 30-year station climatology has the lowest validation MAE, RMSE and highest fold-mean $R^2$.

BUSINESS TOLERANCE

### Best hit rate: 48.48%

The regional February 1 temperature rule has the highest proportion of forecasts within ±3 days.

## Performance across validation folds

| Fold | Validation years | N | Climatology MAE | National Feb MAE | National Mar MAE |
|---|---|---|---|---|---|
| 01 | 2005–06 | 148 | 2.737 | 3.980 | 3.993 |
| 02 | 2007–08 | 127 | 3.383 | 5.795 | 5.827 |
| 03 | 2009–10 | 106 | 3.618 | 4.255 | 4.151 |
| 04 | 2011–12 | 98 | 3.232 | 4.571 | 4.653 |
| 05 | 2013–14 | 96 | 6.049 | 5.490 | 6.250 |

Fold-level national temperature-rule values are reproduced from the executed temperature notebook. The regional model's aggregate metrics are presented above; its per-fold values are not restated here without a verified fold table.

The national-to-regional temperature-rule change reduced average fold MAE from **4.818 to 4.287** days (February 1, approximately **11.0%**) and from **4.975 to 4.383** days (March 1, approximately **11.9%**). Yet the regional March 1 configuration remained slightly less accurate than regional February 1 on MAE.

## Forecast completeness and diagnostics

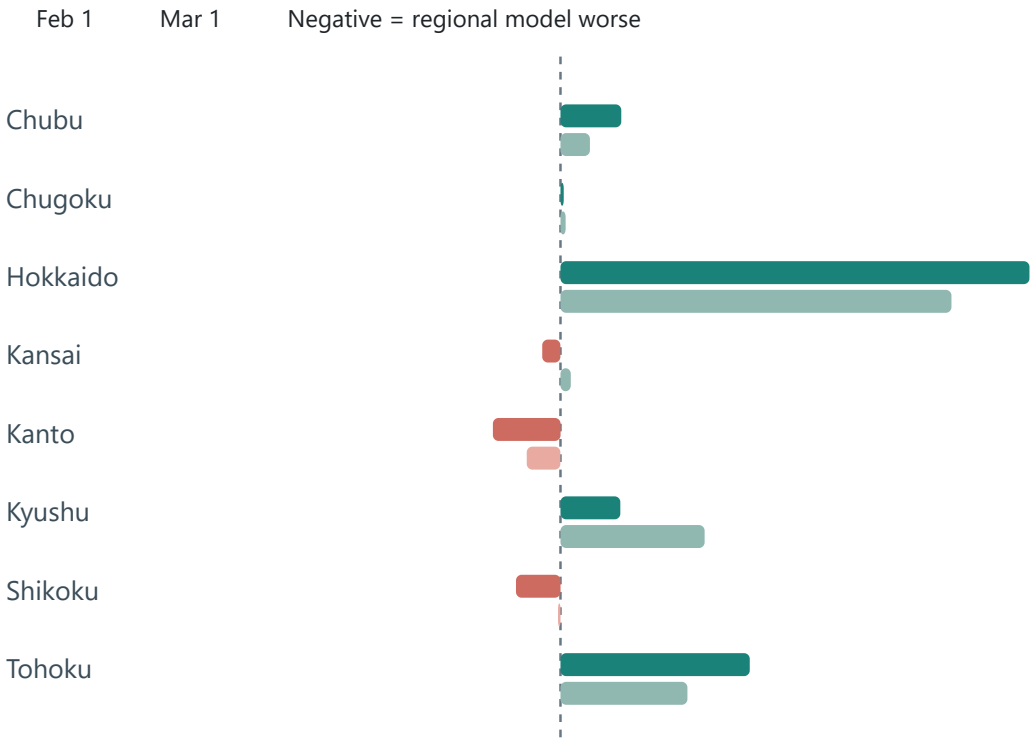
| Check | Observed result |
|---|---|
| National temperature forecasts | 575 / 575 threshold crossings for each issue date |
| Regional temperature forecasts | 575 / 575 threshold crossings for each issue date |
| Regional predictions before issue date | 0 |
| Missing observed-temperature substitutions | 0 in reported national diagnostic |
| National weather-climatology fallback days | 0 in reported national diagnostic |
| National pooled signed error | +1.310 days (Feb 1); +0.503 days (Mar 1) |

06 / GEOGRAPHIC HETEROGENEITY

# Where regional calibration helps — and where it does not

Regional calibration is beneficial overall, but performance varies. The table compares paired national and regional predictions on the same validation station-years.

### MAE reduction by region




Bars to the right of zero indicate reduced MAE from regional calibration; bars to the left indicate deterioration.

| Region | N / issue date | Regional Feb MAE | Regional Mar MAE | Feb improvement | Mar improvement |
|---|---|---|---|---|---|
| Chubu | 110 | 3.591 | 3.700 | +0.600 | +0.291 |
| Chugoku | 61 | 3.115 | 3.311 | +0.033 | +0.050 |
| Hokkaido | 35 | 5.114 | 5.371 | +4.629 | +3.858 |
| Kansai | 89 | 3.371 | 3.663 | -0.180 | +0.101 |
| Kanto | 81 | 5.383 | 5.185 | -0.667 | -0.333 |
| Kyushu | 83 | 6.024 | 5.771 | +0.590 | +1.422 |
| Shikoku | 41 | 3.341 | 3.463 | -0.439 | -0.024 |
| Tohoku | 75 | 3.773 | 4.000 | +1.867 | +1.253 |

### Largest benefits

**Hokkaido:** MAE reduction of 4.629 days (Feb) and 3.857 days (Mar). **Tohoku:** improvements of 1.867 and 1.253 days, respectively.

### Important exceptions

**Kanto and Shikoku:** regional calibration increases MAE for both dates. **Kansai:** slightly worse on February 1, slightly better on March 1.

Regional calibration improves **5 of 8** regions for February 1 and **6 of 8** for March 1. These are descriptive validation findings, not proof that regional calibration will dominate on unseen years.

07 / INTERPRETATION & CAUTION

# Assumptions, limitations and leakage safeguards

### What was enforced in the forecasting code

- Five expanding-window folds and shared station-year evaluation sample.
- Training-fold-only calibration of temperature thresholds and daily temperature climatologies.
- February 1 uses no realized February weather; March 1 uses realized February weather only.
- Observed weather is used strictly *before* the issue date.
- 2015–2025 is reserved as the final held-out test period.

## Known limitations

1 **Historical weather is not a seasonal forecast**
Future temperatures are replaced with historical daily averages. Unusual spring warmth or cold spells cannot be anticipated.

2 **Geographic aggregation remains coarse**
Even eight regional thresholds may hide station-level microclimates, elevation differences and coastal effects.

3 **Long-term change and temporal instability**
Climatology assumes recent historical bloom timing remains informative. Fold 5 shows particularly high error.

4 **Target and predictor availability**
The standardized feature names align with forecast cutoffs, but upstream imputation and feature construction were not independently audited end-to-end.

5 **Calendar and accumulation edge cases**
The code uses a February 28 climatological fallback for February 29. The leap-year convention and exact treatment of incomplete sequences should be verified before deployment.

6 **Validation selection is not independent testing**
National and regional configurations were compared on the same validation years used for model selection. Final generalization remains unmeasured.

**Data-quality example — Onahama 1977:** The weather file showed 35 missing Tmax values over February–June, but the February 1–April 14 interval through observed full bloom contained all 73 expected days with no missing Tmax. Therefore, the station-year remained eligible for historical full-bloom threshold calibration. This does not imply all subsequent weather records are complete.

### Interpretation of $R^2$

A relatively high pooled or fold-level $R^2$ can partly reflect substantial geographic differences in typical bloom dates across Japan. It should not be interpreted as evidence that the model reliably predicts each station's year-to-year bloom anomaly.

### What remains open

**Not yet independently verified:** full upstream weather-imputation cutoff safety; leap-day and missing-data edge cases; deployment readiness. The 2015–2025 held-out test evaluation is complete and reported above.

08 / INDEPENDENT EVALUATION · 2015–2025

# Final held-out baseline testing

The configurations were selected using the 2005–2014 expanding-window validation folds, then fitted on the 1964–2014 development period. Notebook 05 generated and evaluated **528 held-out station-year predictions per model and issue date** for 2015–2025. These are actual executed test results, not validation estimates.

| | | | |
|---|---|---|---|
| **5.040 days** | **37.50%** | **528 / 528** | **0** |

Best test MAE · 30-year climatology

Best test ±3-day hit rate · regional Mar 1

Predictions generated per model/date

Predictions before issue date

## Final test metrics — six model/date combinations

| Model | Issue date | N | MAE (days) ↓ | RMSE (days) ↓ | $R^2$ ↑ | ±3-day hit rate ↑ |
|---|---|---|---|---|---|---|
| 30-year climatology | Feb 1 | 528 | **5.040** | **6.194** | **0.500** | 35.23% |
| Regional temperature rule | Feb 1 | 528 | 5.915 | 7.261 | 0.312 | 34.28% |
| National temperature rule | Feb 1 | 528 | 7.339 | 9.385 | -0.149 | 32.01% |
| 30-year climatology | Mar 1 | 528 | **5.040** | **6.194** | **0.500** | 35.23% |
| Regional temperature rule | Mar 1 | 528 | 5.555 | 6.859 | 0.387 | **37.50%** |
| National temperature rule | Mar 1 | 528 | 7.157 | 9.035 | -0.064 | 29.73% |

Scores from executed `05_baseline_final_testing.ipynb`, evaluated with the team's `evaluate(y_true, y_pred)`. All hit rates shown as percentages; MAE/RMSE in days. $R^2$ is computed over the complete held-out test sample.

## Test MAE by model and forecast date

| | |
|---|---|
| Climatology · Feb 1 | **5.040** |
| Regional rule · Feb 1 | **5.915** |
| National rule · Feb 1 | **7.339** |
| Climatology · Mar 1 | **5.040** |
| Regional rule · Mar 1 | **5.555** |
| National rule · Mar 1 | **7.157** |

Lower is better. Test-set scores are not interchangeable with the equal-weighted fold-average validation metrics elsewhere in this report.

## Lowest average error

**30-year station climatology** had the lowest test MAE (5.040 days), RMSE (6.194 days) and the highest test $R^2$ (0.500). Its predictions are identical on February 1 and March 1 because it uses no current-year weather.

## Best ±3-day tolerance

**Regional temperature rule — March 1** had the highest test hit rate (37.50%), although its MAE (5.555 days) remained higher than climatology. This distinction matters for travel planning.

### Generalization from validation to held-out testing

The validation-fold mean MAE for 30-year climatology was **3.804 days**, versus **5.040 days** on the test sample. The regional rule rose from **4.287 to 5.915 days** for February 1 and **4.383 to 5.555 days** for March 1. This indicates weaker performance on the later test years, but note that the validation figures average five fold-level MAEs equally whereas the test figure is one pooled 2015–2025 MAE. These are informative comparisons, not identical estimands.

Regional calibration continued to outperform the national temperature rule on both test issue dates: MAE was 5.915 vs 7.339 days for February 1 and 5.555 vs 7.157 days for March 1. Unlike the validation result, the **regional March 1** model now outperformed regional February 1 on both MAE and hit rate. These observations should be interpreted as results, not as grounds for post-test retuning.

### Frozen fitted temperature thresholds (1964–2014)

National full-bloom threshold: **746.10°C**. Region-specific medians were fitted from the development data only:

| Region | Threshold (°C) | Region | Threshold (°C) |
|---|---|---|---|
| Chubu | 717.40 | Chugoku | 747.20 |
| Hokkaido | 573.80 | Kansai | 762.35 |
| Kanto | 786.05 | Kyushu | 845.00 |
| Shikoku | 799.20 | Tohoku | 664.70 |

**Audit status:** Notebook 05 reported zero flagged imputed February temperature observations, zero missing Tmax values among the existing test-February rows, complete prediction coverage, and no pre-issue predicted dates. This is not an independent proof that upstream ETL was entirely leakage-free; leap-day and upstream imputation audits remain open. The 2015–2025 test results should not be used to retune the selected baselines.

**08 / RECOMMENDATIONS**

# Conclusions and team handover

### Recommended benchmark set

**Primary historical benchmark:** 30-year station climatology, selected because it has the lowest validation MAE (3.804 days). **Preferred temperature benchmark:** regionally calibrated accumulated-temperature rule, selected because it improves on national calibration for both forecast dates. **Reference:** retain the national threshold version to quantify the value of regional calibration.

For validation, 30-year climatology had the lowest average error while regional February 1 had the highest ±3-day hit rate. In the independent test, climatology still had the lowest MAE, while regional March 1 achieved the highest hit rate. The team should compare its chosen champion models with these frozen baseline test results on the same eligible 2015–2025 observations.

## Handover outputs

| Artifact | Purpose |
|---|---|
| `01_baseline_data_validation.ipynb` | Structural dataset checks and cutoff feature review |
| `02_station_climatology.ipynb` | 30-year historical benchmark and CV results |
| `03_temperature_rule.ipynb` | National/regional threshold calibration and cutoff-aware forecasts |
| `04_baseline_comparison.ipynb` | Metric tables, visual comparisons and regional error analysis |
| `outputs/baseline/*.csv` | CV predictions, fold metrics and paired-error comparisons |

> The central lesson is not that a more complex baseline always wins: a well-defined historical average can be difficult to beat, while geographic calibration meaningfully improves a temperature heuristic without fully solving its limitations.

APPENDIX / REPRODUCIBILITY

# Methodological record

▼ **Evaluation conventions**

MAE and RMSE are expressed in days; hit rate is stored as a proportion internally and displayed as a percentage. The table uses equal weighting across the five folds, even though folds have different numbers of observations (148, 127, 106, 98, 96). The $R^2$ reported is the arithmetic mean of fold $R^2$ values, not pooled $R^2$.

▶ **Calibration and forecasting decisions**

▶ **Source materials and provenance**

▶ **Recommended next project step**

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Sakura Yosoku · Baseline Modelling & Benchmarking · Updated 16 October 2026 | Includes executed held-out test results from Notebook 05; upstream ETL/leap-day audits remain outstanding.